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Usability Heuristics and Explicit Analysis of UI Design for Optimum User Experience: The Case of Pili (Canarium ovatum Engl.) Information System and Marketplace

Salvador V. Briones II
Institute of Information Technology
Partido State University
Goa, Camarines Sur, Philippines
e-mail: salvador.briones@parsu.edu.ph

Ronnel R. Atole
Institute of Information Technology
Partido State University
Goa, Camarines Sur, Philippines
e-mail: ronnel.atole@parsu.edu.ph

Leo Constantine S. Bello
Institute of Information Technology
Partido State University
Goa, Camarines Sur, Philippines
e-mail: lbello@parsu.edu.ph

John Rey S. Lirag
Institute of Information Technology
Partido State University
Goa, Camarines Sur, Philippines
e-mail: johnreylirag@parsu.edu.ph

Rowell John B. Artiaga
Institute of Information Technology
Partido State University
Goa, Camarines Sur, Philippines
e-mail: john.artiaga@parsu.edu.ph

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Abstract— **The intricateness and dynamism of any software development project are viewed as prevailing attributes that affect its outcome. However, explicit utilization of recognized software development principles and standards ensures a higher possibility of achieving greater success. This paper presents the results of the efforts in the design and development of Comprehensive Pili (Canarium ovatum Engl.) Information System and Marketplace particularly the user experiences. This project was funded by the Department of Science and Technology (DOST) with the intention of establishing a dynamic web of knowledge for the Pili industry, thus addressing the dearth of accessible Pili information. The development process follows strict adherence to PiliIS and Marketplace Software Development Paradigm - researchers made software development procedures adopting the Agile software development concepts. This paradigm ensures explicit compliance to ISO/IEC 25010 standards. An internal evaluation team was employed to examine the user interface judging its compliance with the recognized usability principles. A diverse group of stakeholders evaluated the system following Jakob Nielsen's principles for interaction design. Complying to these highly recognized software quality standards and adhering to the researchers' paradigm, the developed software was marked by stakeholders as "highly usable", "highly acceptable" and "highly reliable" with average ratings of 94.65%, 93.71%, and 93.53%, respectively. Additionally, Philippine Pili Industry League, Inc. (PhilPili) rated the system as "excellent". Given the limited study on the development and evaluation of information systems designed to support the marginalized sectors in the country, this study provides significant insights on the software quality principles and competencies as universal standards that boost favorable customer satisfaction and optimum user experience.**

Keywords- Software development, Canarium ovatum Engl., usability heuristics, software evaluation, user experience

I. INTRODUCTION

A. *The Pili Domain*

The Pili (Canarium ovatum Engl.) tree is indigenous in the Philippines and can grow easily in tropical countries [1]. It is one of the major fruit crops that are registered under the National Seed Industry Council (NSIC). Pili trees are abundantly growing in the provinces of Albay, Camarines Norte, Camarines Sur, and Sorsogon and are noted as a "flagship commodity" of the Bicol Region in the Philippines. Hon. Salvador B. Belaro and Hon Gabriel H. Bordado filed House Bill number 6717, noting Pili as one of the important commodities sold by the Bicol Region and viewed to be one of the Philippine products with the greatest advantage and competitiveness on the global scale.

The Pili tree lives for many years and has a very sturdy trunk and branches making it one of the most resistant trees to typhoons. The tree is fairly large that reaches a height of about twenty (20) meters and a trunk diameter of about forty (40) centimeters. Leaves are alternate and compound with opposite smooth leaflets, which are rounded at the base and pointed at the tip. Although some consider it as a minority crop, numerous pharmaceutical and industrial benefits can be gained from this tree. The kernel can be eaten raw or processed and is primarily used as food ingredients. The pulp and the kernel are both producing oil which is used in culinary and considered equal in quality with olive oil and said superior to almond oil [2]. The shells are used as home and body accessories. The trunks produce a resin that has been an export product of the country for years [3]. Moreover, the wood from this tree is characterized by fine striated grains making it ideal for the manufacture of high-quality furniture [4].

Because of its known benefits and uses, the Pili producers, processors, farmers, researchers, and the Pili

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industry in general, considered it as a "tree of hope", "the majestic tree", and "a premium crop" [5]. With the growing number of people settling in the Pili industry, believing the possibilities of high economic potentials, support from different sectors of the country is necessary to enliven and invigorate the stakeholders in this industry.

B. *The Need to Develop a Pili Repository of Knowledge*

Commercialization of pili products, though to some extent has been a success, is yet to be backed by readily accessible data and knowledge. Despite the fact that pili products are everywhere these days, there is no sufficient, comprehensive, and easily retrievable knowledge on this particular industry. It is believed that making information and data on Pili publicly available will shore up and incite the stakeholders in this industry and help boost yield/production, encourage more research and development efforts, promote innovation, step up marketability of various pili-derived products, and consequently improve the living conditions of marginalized farmers and vendors.

To support this undertaking, the DOST through the Niche Centers in the Region for R&D (NICER) provides an institutional grant to higher education institutions in the Regions for R&D capacity building. In Region V, Bicol Region, the Bicol Consortium for Agriculture, Aquatic, and Natural Resources Research and Development (BCAARRD) handles the NICER program, thus titled it as Pili NICER Program. One of the components of the Pili NICER program is the development of a Comprehensive Pili Information System and Marketplace (PiliIS and Marketplace). A web application program that supports the aim of providing and collecting global information about Pili and the Pili industry in general.

The development of the comprehensive PiliIS and Marketplace is viewed as a response to the challenges having identified Pili as a regional priority crop of the Bicol Region. This facility serves as a repository of information to DOST and the Bicol Consortium for Agriculture, Aquatic, and Natural Resources Research and Development (BCAARRD) and its member agencies, producers, processors, traders, researchers, and the Pili industry as a whole. Moreover, since it is a knowledge bank, it helped address the dearth of accessible knowledge on the Pili industry and could be contributory to the development of the industry. The system was designed with flexibility in mind to interface and interconnect/link with other DOST information systems allowing for a dynamic web of knowledge.

Considering the diverse users of the system, with different levels of understanding of information and communications technology (ICT) facilities/tools and equipment, it is imperative that the system produce outputs according to their requirements and guarantee that the system's interface is comprehensible and has a facilitating usability for their satisfaction and optimum user experience.

C. *Instruments Used*

The user interface (UI) is the most important part of any computer system. It allows the user to interact with the system effectively and efficiently. UI includes basic design and style elements while maintaining strong functionality. Good screen design according to Nichols [6], can improve portability, media richness, engagement, and support. To be truly effective, UI requires an understanding of many things, including the characteristics of people and how information must be visually presented to enhance human acceptance and comprehension, thus considering the capabilities and limitations of the hardware and software of the human-computer interface.

The works of Berezhnoi [7] pointed out that UI should not only focus on the aesthetics of the screen but also maximize responsiveness, efficiency, and accessibility of the system. In other words, UI design helps improve the conversion rate of any computer program, particularly online applications. In today's context of software development, especially on the evaluation of usability, Kolling and Mckay [8] takedown "heuristic evaluation" as a practical method of assessing software quality and in identifying potential problems for a given target group or context. Heuristics evaluation for computer programs helps to identify usability problems in the user interface (UI) design. This was popularized by Jakob Nielsen through his ten (10) usability heuristics for user interface design. This method is typically conducted by a group of evaluators inspecting the interface of the developed software, usually, the inspection is done alone to ensure an independent and unbiased result [9].

Furthermore, the usability of the developed software was the decisive objective of the study. Usability as mentioned in the study of Punchooji and Hongwarittorn [10] is the extent to which a product can be used by specified users to achieve specific goals with effectiveness, efficiency, and satisfaction in a specified context of use.

II. METHODOLOGY

A. *Software Quality Evaluation and Analysis*

Different breeds of the meta-level descriptions of software quality evaluation exist. These are often characterized as universal paradigms with a series of various levels of analysis. A well-established quality standard can bring about many advantages to all stakeholders. The entire process of the development of the PiliIS and Marketplace (Figure 1 and Figure 2) follows strict adherence to the software development procedures, the PiliIS and Marketplace Software Development Paradigm (Figure 3). This paradigm is a systematic integration of methods and techniques based on the developers' software development practices and partially adopting concepts and standards from Agile software development methodologies. Specifically, these procedures established explicit compliance on the ISO/IEC 25010 standards [12] (Figure 4) that ensures a functional, efficient, compatible, maintainable, reliable, portable, secure and usable software will be delivered to the target users in due time. During the verification process, the researchers employ an evaluation team that examines the interface and judges its compliance with the recognized software quality principles. Accordingly, this evaluation method helps to identify the usability issues and problems in the user interface design.

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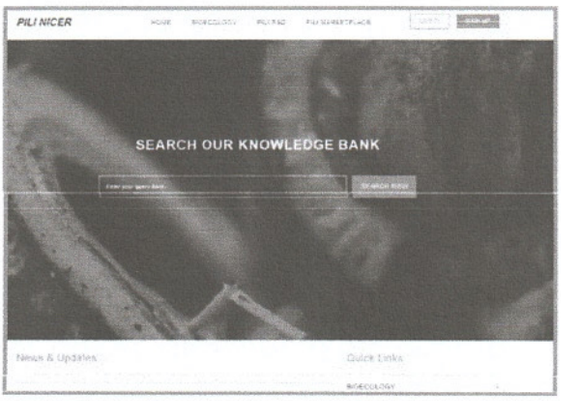


Figure 1. The PiliIS Homepage.

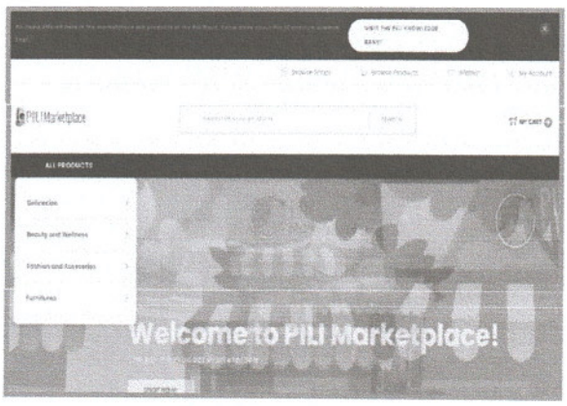


Figure 2. Pili Marketplace Homepage.

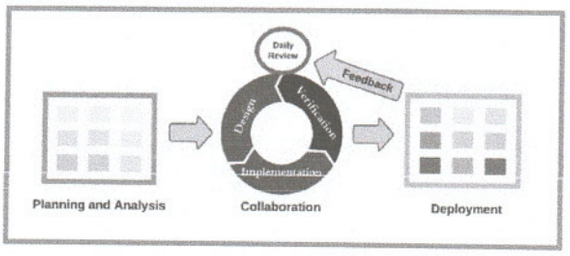


Figure 3. PiliIS and Marketplace Software Development Paradigm.

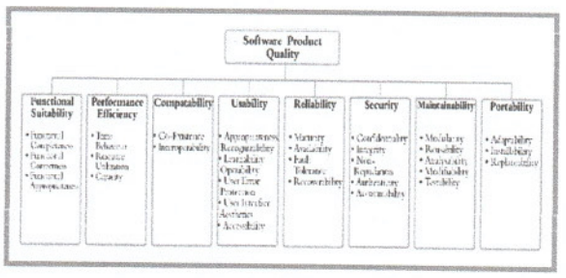


Figure 4. ISO/IEC 25010 Standards

B. *Data Gathering*

A survey instrument was developed to further validate the efficiency and effectiveness of the software output particularly on the aspects of PiliIS and Marketplace's acceptability and reliability. The developed software was further subjected to testing and evaluation by a diverse group of stakeholders following Jakob Nielsen's Usability Heuristics for User Interface Design (Figure 4). Specifically, this process measures the conformance of the user interface to the standards of UI design, further identifies any usability problems, collects qualitative and quantitative data, and determines the stakeholder's overall satisfaction with the system. In other situations, interviews and focused group discussions with the stakeholders were conducted such as in formulating the usage policy for the Pili IS and Marketplace operation, and in validating secondary data.

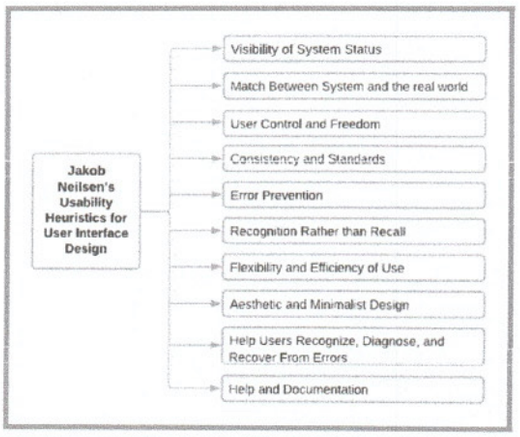


Figure 5. Jakob Nielsen's Usability Heuristics for User Interface Design

C. *Stakeholders of the Study*

The major stakeholders of this study are the farmers, producers, processors, vendors, and policy-makers who are engaged in Pili production, processing, and trading and members of PhilPili. Other stakeholders of this study are members of BCAARRD, representatives from various government agencies in Region V, such as Agricultural Training Institute - Regional Training Center (ATI-RTC), Department of Agriculture (DA), DOST, Department of Trade and Industry (DTI), Provincial Government of Albay, Provincial Government of Sorsogon, Philippine Coconut Authority - Albay Research Center (PCS-ARC), and the researchers from the different academic institution in Region V such as Ateneo De Naga University, Bicol State College of Applied of Sciences and Technology, Bicol University, Camarines Norte State College, Camarines Sur Polytechnic Colleges, Catanduanes State University, Central Bicol State University of Agriculture, Partido State University, and the Sorsogon State College.

III. RESULTS AND DISCUSSION

Experiences in software development and the use of defined and recognized software development methodology helped immensely in arriving at highly facilitating and usable software output. The entire process of the development of PiliIS and Marketplace follows strict adherence to the PiliIS and Marketplace Software Development Paradigm. This paradigm is essentially a systematic integration of methods and techniques that

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sprung from the developers' software development experiences and practices, relatively adapting the Agile software development concepts.

The PiliIS and Marketplace Software Development Paradigm focus on a disciplined software development environment that encourages accountability and collaboration between the members of the team. The project primarily started through a systematic planning and analysis phase which is essentially a high level feasibility discussion of the objectives, vision, and resources for the project. The project's requirements were scrutinously defined and the return on investment (ROI) was properly justified. The results of it were conveyed to the collaboration phase. This phase marks the start of the first iteration of the software project with the aim of delivering a working version in its due time. Daily reviews were done to ensure the efficiency and effectiveness of the conducted activities and the productivity of the team. A verification process was also employed to examine the interface and judge its compliance with the recognized software quality principles. Accordingly, this evaluation method helps to identify the usability issues and problems in the user interface design.

The final process in the paradigm was the deployment of the system. The system was closely monitored for bugs and some defects that may have been missed during the verification phase. During this stage, relevant training with the stakeholders was conducted to ensure a highly usable, highly reliable, and highly acceptable software product will be handed over to them.

This paradigm guarantees that the developed software is explicitly compliant with the ISO/IEC 25010 standards, a model for evaluating software product quality. The parameters that define and measure the quality of the software and the extent to which this software satisfies the stated and implied needs of the stakeholders are functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

The developed PiliIS and Marketplace were further subjected to testing and evaluation by the diverse group of stakeholders following Jakob Nielsen's [9] principles for interaction design, specifically measuring the conformance of the user interface to the standards of UI design. The result of this evaluation is "highly usable" with an average rating of ninety-four and sixty-five hundredths percent (94.65%), as presented in Table 1, the usability percentage of PiliIS and Marketplace.

TABLE I. USABILITY PERCENTAGE OF PILIIS AND MARKETPLACE

No.	Usability Heuristics Indicators	Summary of Results (%)
1	Visibility of System Status	95.45
2	Match Between System and the Real World	94.55
3	User Control and Freedom Support undo and redo.	96.36
4	Consistency and Standards	94.55
5	Error Prevention	92.6
6	Recognition Rather than Recall	94.24
7	Flexibility and Efficiency of Use	94.85
8	Aesthetic and Minimalist Design	94.85
9	Help Users Recognize, Diagnose, and Recover from Errors	94.85
10	Help and Documentation	94.24
	TOTAL	**94.65**

[a] **Note:** 90-100, Highly Usable; 70-89, Usable; 50-69, Fairly Usable; 30-49, Less Usable; 29-0, Unusable

By firmly complying with these known and highly recognized software quality standards and decisively adhering to the paradigm, the stakeholders marked the PiliIS and Marketplace as "highly acceptable" and "highly reliable" with the average ratings of ninety-three and seventy-one hundredths percent (93.71%) and ninety-three and fifty-three hundredths percent (93.53%), respectively.

Representatives from the esteemed PhilPili - a non-profit organization engaged in proper utilization, regulation, and policy-making for the Pili industry, rated the system "excellent", the highest rating the stakeholder can give. Table 2 and Table 3 presents the summary of ratings on acceptability and reliability.

TABLE II. ACCEPTABILITY PERCENTAGE OF PILIIS AND MARKETPLACE

No.	Acceptability Indicators	Summary of Results (%)
1	The objectives of the PiliIS and Marketplace were clearly defined.	94.24
2	The PiliIS and Marketplace are useful at my present work.	94.55
3	The use of PiliIS and Marketplace facilitates my tasks.	96.06
4	Information provided in PiliIS and Marketplace are helpful.	93.64
5	The contents are well organized and consistent with the objectives of the PiliIS..	92.73
6	Participation and interaction of users were encouraged.	94.55
7	Instructions were clear and understandable.	94.24
8	PiliIS and Marketplace provided opportunities to practice and reinforce previous knowledge.	93.03
9	Users are able to contribute contents into the information repository through their accounts.	91.52
10	I will use the PiliIS and Marketplace to assist my operation.	92.6
	TOTAL	**93.71**

[a] **Note:** 90-100, Highly Acceptable; 70-89, Acceptable; 50-69, Fairly Acceptable c; 30-49, Less Acceptable; 29-0, Unacceptable

TABLE III. RELIABILITY PERCENTAGE ON PILIIS AND MARKETPLACE

No.	Reliability Indicators	Summary of Results (%)
1	The information presented in the system is accurate, complete and current.	95.15
2	Content and context are consistent with the objectives of the PiliIS.	95.45
3	All information relates to the stated purpose and project goals.	96.67
4	Multimedia appears to be directly related to stated purpose and project goals.	95.15
5	Layout is clear and intuitive, users can always find what they need.	91.82
6	Pages works in most common browsers like Mozilla, Google Chrome, Internet Explorer, etc.	95.76
7	Paragraphs and sections have clear and accurate informative headings.	92.12

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8	Clear and complete directions are available for access.	92.12
9	Help users recognize, diagnose, and recover from errors.	88.48
10	Can bring a failed system to full operation.	92.6
	TOTAL	**93.53**

[a] Note: 90-100, Highly Reliable; 70-89, Reliable; 50-69, Fairly Reliable; 30-49, Less Reliable; 29-0, Unreliable

IV. CONCLUSION AND RECOMMENDATIONS

In any software development project, the intricateness and dynamism of the software are viewed as the prevailing attributes that affect the outcome of the project. However, with the explicit utilization of the recognized software development practices and the acknowledged paradigms along with the clear vision and right attitude towards the development of the project, there is likely a high possibility of achieving greater success for the project.

Presented in this paper are the results of the efforts in the design and development of the Comprehensive Pili (Canarium ovatum Engl.) Information System and Marketplace, the user experiences, and the stakeholder's response on the usability and UI design of the system. By firmly complying with the known and highly recognized software quality standards and decisively adhering to the PiliIS and Marketplace Software Development Paradigm, the stakeholders mark the PiliIS and Marketplace as "highly usable", "highly acceptable" and "highly reliable".

Given the limited study on the development and evaluation of information systems designed to support the marginalized sectors in the country, it is our hope that this research study provides significant insights on the software quality principles and competencies as universal standards for various software development projects that boost favorable customer satisfaction and optimum user experience.

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REFERENCES

[1] L. J. Pham, and N. Dumandan, "Philippine Pili: Composition of the Lipid Molecular Species", Journal of Ethnic Food, Vol 2, Issue 4, 2015, Science Direct. Retrieved from https://www.sciencedirect.com/science/article/pii/S2352618115000621

[2] Briones, M. R. SA. (N.D.). The promising of Pili. Bureau of Agricultural Research. Retrieved from https://bar.gov.ph/index.php/test-archive/419-may-2002-issue/3611-the-promising-pili

[3] Belaro, S. B. and Bordado, G. H. Jr. (2017). House Bill No. 6717, "An Act Promoting the Cultivation, Production, Processing, Marketing and Distribution of Pili Nut and for Other Purposes", Seventeenth Congress, House of Representatives, Philippines. Retrieved on from https://www.congress.gov.ph/legisdocs/basic_17/HB06717.pdf

[4] Rontas, D.Y. (N.D.). The Promising Pili. Bureau of Agricultural Research. Retrieved on March 17, 2021 from https://bar.gov.ph/index.php/test-archive/419-may-2002-issue/3611-the-promising-pili

[5] Imperial, R. M. (N. D.). Pili (Canarium ovatum Engl.) A Promising Indigenous Crop of the Bicol Region in the Philippines for Food and Nutrition. Food and Agriculture Organization of the United Nations. Retrieved from http://www.fao.org/fileadmin/templates/rap/files/meetings/2012/120531_iib3.pdf

[6] Nichols, M. (2020). Reading and Studying on the Screen: An Overview of Literature Towards Good Learning Design Practice. Journal of Open, Flexible Distance Learning. Retrieved from https://search.informit.org/doi/abs/10.3316/informit.195571684952519

[7] Berezhnoi, R. (2019). What is UI design and Why is it important? F5 Studio. Retrieved from https://f5-studio.com/articles/what-is-user-interface-design-and-why-is-it-important

[8] Kolling, M. and Mckay, F. (2016). Heuristic Evaluation for Novice Programming. ACM Transactions on Computing Education, Vol. 16. No. 3. Retrieved from https://dl.acm.org/doi/10.1145/2872521

[9] Neilsen, J. (2020). 10 Usability Heuristics for User Interface Design. NN/g Neilsen Norman Group. Retrieved from https://www.nngroup.com/articles/ten-usability-heuristics/

[10] Punchooji, L and Hongwarittorn, N. (2017). Usability Studies on Mobile User Interface Design Patterns; A Systematic Literature Review. Retrieved from DOI: 10.1155/2017/6787504

[11] Lah, U., Lewis, J. R. and Šumak, B. (2020). Perceived Usability and the Modified Technology Acceptance Model. International Journal of Human Computer Interaction, Vol. 36 2020 – Issue 13. Retrieved from https://www.tandfonline.com/doi/abs/10.1080/10447318.2020.1727262

[12] ISO 25000, (N.D.) ISO/IEC 25010. Retrieved from https://iso25000.com/index.php/en/iso-25000-standards/iso-25010

[13] Jimenez, C., Lozada, P. and Rosas P. (2016). Usability Heuristics: A Systematic Review. 2016 IEEE11th Colombian Computing Conference. DOI: 10.1109/ColumbianCC.2016.7750805

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